

FINAL EXPLANATION: MATHEMATICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: MATHEMATICS GRADE 10: REAL NUMBERS FINAL EXPLANATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

You have finished the unit on real numbers. Now answer the Driving Question: How do we use real numbers in everyday life, and why are reciprocals important? Use Achieng's trip to the duka near Kibuye Market in Kisumu as your main example. Write your answer in the space under each prompt. Use evidence and numbers from your lessons, your Driving Question Board (DQB) and your unit model. Show your working where you calculate. Write clearly and explain your reasoning in full sentences.

Part 1: Different numbers at the duka

Achieng bought 3 items at Ksh 45 each, paid with one Ksh 200 note, bought $\frac{1}{2}$ kg of sugar at Ksh 160 per kg, and saw a tea-leaf sachet tagged Ksh 22.50. (a) Work out her total for the 3 items and her change. (b) Name the type of number used for counting items, for part of a kilo, and for the price tag, and explain why one trip needs all three types. (c) Explain how a fraction and a decimal can name the same quantity, using one example from the trip.

Write your answer here:

Part 2: Classifying whole numbers

Some whole numbers from Achieng's trip are 3, 6, 15, 45, 65, 90, 135 and 200. (a) Classify 3, 15 and 90 as odd or even, and as prime or composite, and justify each using factors. (b) Explain why 1 is neither prime nor composite, and why 2 is special. (c) Seventeen cobs of maize are to be shared equally among groups. What happens, and what does this tell you about 17? (d) Is every odd number prime? Use the duka numbers to support your answer.

Write your answer here:

Part 3: Rational and irrational numbers

(a) Use division to explain what happens to the decimals of $\frac{1}{4}$ and $\frac{1}{7}$, and say whether each is rational. **(b)** What is the definition of a rational number? Show that the whole number 3 and Ksh 22.50 are rational. **(c)** Squaring 1.414, 1.4142 and 1.415 on a calculator gave 1.999396, 1.99996164 and 2.002225. Using these results and the square plot of area 2 m^2 , explain why $\sqrt{2}$ goes on forever and what type of number it is. **(d)** Describe how rational and irrational numbers together make up the real numbers, and place 3, $\frac{1}{2}$, 22.50, $\sqrt{2}$ and π on this map.

Write your answer here:

Part 4: Reciprocals and the tools to find them

(a) What is the reciprocal of a number n , and what do you get when you multiply n by its reciprocal? Which number has no reciprocal? (b) Find the reciprocal of 8, of 2.5 and of 6 (give 6 to 4 decimal places) and check one of them by multiplying. (c) Mzee Odhiambo says 6 mandazi cost Ksh 90, so the unit price is Ksh 15. Explain how this is a reciprocal calculation. (d) Compare the answers for $1/7$ from mathematical tables and from the calculator $1/x$ key. Why do they differ, and are both useful for money?

Write your answer here:

Part 5: Answering the Driving Question

(a) A bus travels from Nairobi to Mombasa, 480 km. Use $\text{time} = \text{distance} \times 1/\text{speed}$ to find the time at 60 km/h and at 80 km/h. (b) Wanjiru saves Ksh 10 000 in a Sacco at 8% simple interest per year. Find the interest for 1 year and for 6 months, then use a reciprocal to find how long she needs to earn Ksh 2 400. (c) Write a final answer to the Driving Question: How do we use real numbers in everyday life, and why are reciprocals important? Connect it back to Achieng's trip.

Write your answer here:

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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Number types and classification	Correctly names whole numbers, fractions, decimals, rational and irrational numbers, and classifies the duka numbers as odd, even, prime or composite using factors, with correct handling of 1 and 2.	Names most number types and classifies most duka numbers correctly, with some factor reasoning.	Confuses number types or classifies numbers without using factors.
Evidence for rational and irrational numbers	Uses the decimal conversions and the squaring checks (1.999396, 1.99996164, 2.002225) to explain clearly why $\sqrt{2}$ is irrational and places numbers on the real-number map.	Uses some evidence and states that $\sqrt{2}$ never ends or repeats, but the link to the squaring checks is partial.	States that $\sqrt{2}$ is irrational without evidence, or confuses recurring and non-recurring decimals.
Reciprocals and tools	Defines the reciprocal, finds correct values, checks with $n \times 1/n = 1$, links to	Defines the reciprocal and finds most values correctly, with limited explanation	Confuses reciprocal with another idea or makes calculation errors with no check.

	unit price, and explains why tables and the calculator differ.	of tools or accuracy.	
Application and accuracy of calculation	All calculations are correct and shown (135, 65, 15, 8 h, 6 h, Ksh 800, Ksh 400, 3 years) using reciprocals.	Most calculations are correct with some working shown.	Several calculation errors or no working shown.
Answer to the Driving Question	Gives a connected answer tying Achieng's trip to real numbers and to reciprocals turning division into multiplication, using examples from several lessons.	Answers the question using the duka trip but gives few links to other lessons.	Gives a list of facts without answering the question.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

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